

Hossain Muhammad Muctadir, EngD. PhD.

✉ muctadir.prof@gmail.com

🌐 <http://muctadir.github.io/>

in muctadir

🐦 @HossainMuctadir

🎓 Google Scholar



Industrial Experience

2025 – Now

📌 R&D Professional

Canon Production Printing, The Netherlands.

As a software engineer and researcher at the R&D unit, I am developing various tools to facilitate the development of high-throughput industrial printing systems.

2018 – 2020

📌 Technological designer in training (part of EngD degree)

Eindhoven University of Technology, The Netherlands.

*During the EngD program, I contributed to several projects. First, in collaboration with **Philips Research**, we integrated AI into their remote healthcare services. My responsibilities included maintaining user stories and overseeing development. Second, our team developed software and mechanical systems for **agricultural drones**, enabling them to autonomously measure soil quality at specified locations. I managed the project, handled scoping, and communicated with stakeholders. Third, in partnership with **ASML**, we created a tool for managing software project dependencies and scope within an IDE. My role encompassed system design and implementation. Finally, for my thesis, I worked with **Unilever** to develop a micro-service based platform for sharing computational models across organisations.*

2017 – 2018

📌 Software Engineer

Drops Technologies, Germany.

I was involved in a variety of tasks, including web services and web application development using .NET MVC, WebApi2, and Angular. Additionally, I implemented a mixed reality Hololens application using Unity3D, created a WebRTC-based video chat application, and organized development activities. I was also responsible for student developer recruitment, supervision, and team activities.

2015 – 2016

📌 Student Software Engineer

FEV GmbH, Germany.

*I implemented FEV-specific plugins of the dSpace Synect and various task automation tools using Python. I also did my **master thesis** here and developed an extensible and modular framework for analyzing similarities among software components and generated reports showing similarity results to help software development teams maintain software product lines with minimal effort.*

2012 – 2014

📌 Software Engineer

Streams Tech Limited, Bangladesh.

I analyzed, designed, and developed web and desktop applications. Projects included creating a software project management web app using C#, ASP.NET MVC, and AngularJS, and a desktop app for generating digital graph data from scanned images using C# .NET and Windows Forms.

Education

2020 – 2025

📌 Doctor of Philosophy (PhD) in Model-Based Software Engineering

Eindhoven University of Technology, The Netherlands.

Thesis title: Consistency matters: Building consistent Digital Twin virtual entities.

2018 – 2020

📌 Engineering Doctorate (EngD) in Software Technology

Eindhoven University of Technology, The Netherlands.

Thesis title: Model as a Service : Towards a Discovery Platform for Internet of Food.

Conducted in collaboration with Unilever R&D, Wageningen, The Netherlands

Education (continued)

- 2014 – 2017  **Masters** in Software Systems Engineering
RWTH Aachen University, Germany.
Thesis title: *Similarity analysis framework for Software Product Line extraction*.
Conducted in collaboration with FEV GmbH, Alsdorf, Germany
- 2009 – 2013  **Bachelors** in Information Technology (major in Software Engineering)
Institute of Information Technology, University of Dhaka, Bangladesh
Thesis title: *Chest x-ray analysis to detect abnormal mass tissue in lung*.

Skills and expertise

- Core Skills  Strong problem-solving and software engineering skills with a focus on model-driven software engineering, object-oriented software development, software design and architecture, and web application development.
- Programming  Python, C#, Java, JavaScript, MATLAB, $\LaTeX$
- Databases  MySQL, PostgreSQL, SQLite, Neo4j, ClickHouse
- Languages  Fluent in spoken and written *English*. Native *Bengali* speaker. Elementary proficiency in *Dutch* and *German*.

Academic experiences and activities

Teaching at Eindhoven University of Technology

- 2024  Instructor at 2nd year computer science bachelor's course Software Design (2IRR00)
- 2023  Supervised a group of bachelor students for final Software Engineering Project (2IPE0)
- 2021 – 2023  Instructor at 2nd year computer science bachelor's course Programming Methods (2IPC0)

Workshops, talks, and research visits

- 2025  Presented our SoSyM journal paper titled "Current trends in digital twin development, maintenance, and operation: An interview study" at the *SE2025 conference at Karlsruhe, Germany*
- 2024  Participated in the workshop on *Digital Twins for Disaster Early Warning Systems* organised by *Netherlands eScience Center* at *CWI, Amsterdam*
-  Received an invitation to present our SoSyM journal paper titled "Current trends in digital twin development, maintenance, and operation: An interview study" at the *Journal First track of MODELS 2024 conference at Linz, Austria*
- 2023  Participated in *19th CAMPaM workshop* on multi-paradigm modeling, model-driven engineering, digital twins at *Bellairs Research Institute, Barbados*
-  Presented about our interview study on current trends in digital twin development, maintenance, and operation at *VERSEN (Dutch organisation of software engineering research) symposium*
- 2022  Participated in *8th NEMO summer school* focusing on the design and implementation of Enterprise Digital Twins and Ecosystems based on conceptual modelling methods at *University of Vienna, Austria*
-  Participated in *MDENet Annual Symposium* on model-driven engineering at *Kings College London, UK*
-  Participated in *IPA (Dutch institute of programming research and algorithmics) Fall Days* on Scalable and Versatile Software Analytics: ML and NLP to the Fore

Academic experiences and activities (continued)

- 2021  Presented about inconsistency detection among cross-domain digital twin models at *ICT Open, yearly Dutch national ICT symposium*

Awards

- 2024  Best paper award at *SEAA 2024 conference* for the paper "Maintaining consistency of digital twin models: exploring the potential of graph-based approaches" at *Paris, France*
- 2022  MDENet travel funding to join the *MDENet annual symposium at Kings College London, UK*

Services

- 2025  Program committee member at ICM (co-located with STAF 2025) and MoM (co-located with MODELS 2025) workshops.
- 2020 – 2024  As a sub-reviewer, reviewed several scientific articles submitted to MODELS, EDTconf, and SPLC.
- 2018 – 2020  As a board member of ATMOS (association of EngD students at TU Eindhoven), I co-organized various events.

Student Supervision

- 2025  Emre Kenar. *An Object-Oriented Language-Based Approach for Stream Reasoning*
Master thesis at TU Eindhoven co-supervised with Loek Cleophas
-  Yanyifan Liao. *Consistency analysis of models used in a brewery digital twin*
Master thesis at TU Eindhoven co-supervised with Loek Cleophas
- 2023  Judith Houdijk. *Taxonomizing and Analyzing Change Impact in Language Oriented Programming*
Master thesis at TU Eindhoven co-supervised with Loek Cleophas and Andreas Wortmann
-  Xingyu Liu. *Digital twin based anomaly detection of Turtle soccer robots*
Master thesis at TU Eindhoven co-supervised with Mark van den Brand
-  Mohammad Ibrahim. *A Graph Database Design for Multi-Domain Model Management*
Engineering Doctorate at TU Eindhoven co-supervised with Mark van den Brand
- 2022  Gijs Walravens. *Towards Digital Twins for soccer robots: a use case in reusing artifacts.*
Master thesis at TU Eindhoven co-supervised with Loek Cleophas

Research Publications

- Muctadir, H. M.**, Kamburjan, E., Cleophas, L., & van den Brand, M. (2026). A consistency management framework for digital twin models. *Journal of Systems and Software*, 234, 112750.
 <https://doi.org/10.1016/j.jss.2025.112750>
- Muctadir, H. M.** (2025, October). *Consistency matters: Building consistent digital twin virtual entities* [Phd Thesis]. Mathematics and Computer Science. Eindhoven University of Technology.
 <https://research.tue.nl/en/publications/consistency-matters-building-consistent-digital-twin-virtual-enti>
- Muctadir, H. M.**, Pfeiffer, J., Houdijk, J., Cleophas, L., & Wortmann, A. A Taxonomy of Change Types for Textual DSL Grammars. In: *In Proceedings of the 13th international conference on model-based software and systems engineering - modelsward*. INSTICC. SciTePress, 2025, 169–176.
 <https://doi.org/10.5220/0013127800003896>

- 4 **Muctadir, H. M.**, Cleophas, L., & van den Brand, M. (2024). Maintaining consistency of digital twin models: Exploring the potential of graph-based approaches. *2024 50th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*, 152–159.
<https://doi.org/10.1109/SEAA64295.2024.00031>
- 5 **Muctadir, H. M.**, Manrique Negrin, D. A., Gunasekaran, R., Cleophas, L., van den Brand, M., & Haverkort, B. R. (2024). Current trends in digital twin development, maintenance, and operation: An interview study. *Software and Systems Modeling*, 23(5), 1275–1305.
<https://doi.org/10.1007/s10270-024-01167-z>
- 6 Bogachenkova, V., Martins, E. C., Jansen, J., Olteniceanu, A.-M., Henkemans, B., Lavin, C., Nguyen, L., Bradley, T., Fürst, V., **Muctadir, H. M.**, van den Brand, M., Cleophas, L., & Serebrenik, A. (2023). Lama: A thematic labelling web application. *Journal of Open Source Software*, 8(85), 5135.
<https://doi.org/10.21105/joss.05135>
- 7 Liu, X., van den Brand, M., **Muctadir, H. M.**, & van de Molengraft, R. (2023). ML-based digital twin for anomaly detection: A case-study on turtle soccer robots. *2023 49th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*, 179–182.
<https://doi.org/10.1109/SEAA60479.2023.00035>
- 8 **Muctadir, H. M.**, König, L., Weber, T., Amrani, M., & Cleophas, L. (2023). Co-evolving meta-models and view types in view-based development. *2023 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, 954–963.
<https://doi.org/10.1109/MODELS-C59198.2023.00150>
- 9 Walravens, G., **Muctadir, H. M.**, & Cleophas, L. (2022). Virtual soccer champions: A case study on artifact reuse in soccer robot digital twin construction. *Proceedings of the 25th International Conference on Model Driven Engineering Languages and Systems: Companion Proceedings*, 463–467.
<https://doi.org/10.1145/3550356.3561586>
- 10 van den Brand, M., Cleophas, L., Gunasekaran, R., Haverkort, B., Negrin, D. A. M., & **Muctadir, H. M.** (2021). Models meet data: Challenges to create virtual entities for digital twins. *2021 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion (MODELS-C)*, 225–228. <https://doi.org/10.1109/MODELS-C53483.2021.00039>
- 11 **Muctadir, H. M.** (2020, October). *Model as a service : Towards a discovery platform for internet of food* [EngD thesis]. Mathematics and Computer Science. Eindhoven University of Technology.
https://pure.tue.nl/ws/portalfiles/portal/163366772/2020_10_15_ST_Muctadir_HM.pdf
- 12 Dey, E. K., & **Muctadir, H. M.** (2014). Chest x-ray analysis to detect mass tissue in lung. *2014 International Conference on Informatics, Electronics & Vision (ICIEV)*, 1–5.
<https://doi.org/10.1109/ICIEV.2014.6850788>

References

Available on Request